Student-Scale Persistent 3D World Modeling on MOVi-A

Memory-Conditioned ConvGRU with Uncertainty-Gated Writes

CS 599 G1 – Final Project Report

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Abstract

Long-horizon visual prediction in 3D scenes is fundamentally limited by what a model can remember about previously observed views. Recent work argues that maintaining an explicit, geometrically grounded persistent memory is essential for view-consistent forecasting, but reproducing the underlying mechanism at full production scale is impractical in a course-project setting. We therefore study a *student-scale* replication of the persistent-memory mechanism on the MOVi-A benchmark and ask whether the memory pathway alone yields measurable gains under matched training budgets. We instantiate a three-stage ConvGRU pipeline: (i) a no-memory baseline that predicts future frames from context only, (ii) a memory-conditioned model augmented with a rendered persistent 3D voxel memory constructed from context observations, and (iii) a memory-conditioned model with a heteroscedastic uncertainty head that gates memory writes. Evaluated under a unified harness on 320 validation windows from a 200-clip MOVi-A subset, the memory-conditioned ConvGRU reduces validation L1 from 0.0214 to 0.0188 (-12.1%), reduces dynamic-region L1 from 0.1023 to 0.0890 (-13.0%), and improves PSNR from 25.57 dB to 26.68 dB relative to the no-memory baseline at the same number of epochs. Adding uncertainty-gated writes further reduces memory-covered L1 from 0.0577 to 0.0443 (-23.2%) while matching or improving every other metric, and the predicted confidence is positively correlated with per-pixel error (r=0.150). A conditional video-diffusion variant of the same pipeline is documented for completeness; we argue on methodological grounds that a deterministic recurrent backbone provides a cleaner isolation of the memory pathway, and we therefore treat diffusion as an exploratory complement rather than a primary result.

1 Introduction

Predicting future views of a 3D scene requires a model to remember not only what it observed in the immediate past but *where* in the world those observations were located. Frame-local predictors, and even short-context recurrent predictors, degrade rapidly under viewpoint change, occlusion, and the accumulation of prediction error over time. Recent work on *persistent 3D world models* [1] addresses this limitation by maintaining an explicit, geometrically grounded memory of the scene from which target views can be queried.

Because such models are typically trained at scales well beyond what is practical in a one-semester graduate course, our objective is not full reproduction but a controlled replication of the underlying mechanism at student scale. Concretely, we ask:

> *On a small MOVi-A subset and under matched training budgets, does adding an explicit persistent memory improve a recurrent next-view predictor over a no-memory baseline,*

and does a per-pixel uncertainty estimate further increase the reliability of the persistent memory?

We instantiate this question through a compact ConvGRU-based pipeline that runs end-to-end on a single workstation-class GPU. Our contributions are:

1. **Reproducible pipeline.** An end-to-end implementation from raw MOVi clip export through trained checkpoints, validation metrics, and qualitative comparison artifacts, executable from local scripts without cluster access.
2. **Controlled comparison.** A side-by-side evaluation of a no-memory ConvGRU, a memory-conditioned ConvGRU, and an uncertainty-gated memory-conditioned ConvGRU under matched data splits, optimisation settings, and training budgets.
3. **Slice-level evaluation harness.** A unified evaluation procedure reporting overall validation metrics and four diagnostic slices: high motion, occlusion recovery, high memory coverage, and depth-edge-heavy windows.
4. **Exploratory diffusion variant.** An end-to-end documented conditional video-diffusion variant, contrasted with the recurrent backbone in Section 4 on methodological grounds.

2 Related Work

Persistent 3D world models. Recent work formulates world modeling not just as next-frame prediction but as maintaining a persistent 3D representation that observations are written into and future views are queried from [1]. This is the conceptual starting point for our pipeline.

Recurrent video prediction. Convolutional recurrent architectures, such as ConvLSTM [3] and ConvGRU [4], are a standard backbone for spatiotemporal prediction. We use a ConvGRU because it provides a deterministic, single-pass spatiotemporal hidden state that composes cleanly with the persistent memory mechanism we are studying (Section 4).

Diffusion video models. Conditional video diffusion is the state-of-the-art family for high-fidelity generative video prediction [5]. We treat it as an exploratory complement to our pipeline rather than the primary backbone, for methodological reasons explained in Section 4.

Uncertainty-aware prediction. Heteroscedastic uncertainty heads [6] provide a way to predict per-pixel confidence alongside the prediction itself. We use this idea both as an auxiliary loss and as a gating signal for what the model writes into its persistent memory.

Dataset. We use the MOVi-A subset of the Kubric [2] synthetic benchmark, which provides multi-view RGB-D videos with ground-truth camera poses and intrinsics — ideal for studying memory-conditioned prediction at small scale.

3 Task and Notation

Let a clip provide RGB frames $\{I_t\}_{t=1}^{T}$, depth maps $\{D_t\}_{t=1}^{T}$, camera poses $\{P_t\}_{t=1}^{T}$, and intrinsics K. We split each clip into windows of length $T_c + T_p$ where the first T_c frames form the *context* and the next T_p frames form the *target* that the model must predict.

The model receives the context RGB $\{I_t\}_{t=1}^{T_c}$, the context poses $\{P_t\}_{t=1}^{T_c}$, and the target poses $\{P_t\}_{t=T_c+1}^{T_c+T_p}$, and outputs predicted RGB $\{\hat{I}_t\}_{t=T_c+1}^{T_c+T_p}$. Memory-conditioned variants also receive a rendered memory tensor (Section 5.4).

In all our final runs we use $T_c = T_p = 4$ and an image resolution of 64×64.

4 Backbone Selection Rationale

The reference work [1] introduces a specific *mechanism* – writing observations into a persistent 3D memory and querying that memory from novel viewpoints – and attributes long-horizon view consistency to this mechanism rather than to the underlying generative model. A faithful replication of that claim therefore requires a backbone that allows the memory pathway to be isolated from confounding sources of variation. We adopt a ConvGRU recurrent backbone for four complementary reasons.

Alignment with the memory mechanism. Persistent-memory writes are naturally expressed as per-timestep updates over a context sequence, and queries from new views as per-target-frame reads. ConvGRUs [4] provide a canonical spatiotemporal recurrent backbone whose hidden state evolves at the same temporal granularity, which permits the rendered memory condition to be injected directly at the per-frame fusion stage (Section 5.4) while leaving the remainder of the architecture identical to the no-memory baseline.

Ablatability of the memory pathway. Since the no-memory and memory-conditioned models share encoder, recurrent cell, and RGB decoder, the memory model can be warm-started from the converged no-memory checkpoint, and the memory pathway becomes a single, well-defined modification. Any change in validation L1, dynamic L1, or memory-covered L1 between the two configurations is therefore directly attributable to the memory pathway – which is precisely the claim under test.

Deterministic, single-pass evaluation. ConvGRU produces deterministic predictions in one forward pass, so the reported metrics (L1, dynamic L1, memory-covered L1, PSNR) characterise the prediction itself rather than a sampling distribution. Generative backbones such as conditional video diffusion produce a distribution of plausible samples per query and require multi-step denoising at evaluation time, which introduces additional variance and complicates attribution of metric changes to the memory pathway as opposed to the sampler.

Compositionality with per-pixel uncertainty. The uncertainty extension (Section 5.5) employs a heteroscedastic head [6] that predicts a per-pixel log-variance over the same hidden features used for prediction. This composes naturally with a deterministic backbone: the resulting confidence map gates memory updates at the same granularity at which the memory is queried. Combining a heteroscedastic head with a denoising diffusion sampler would entangle aleatoric uncertainty with sampler-induced variability, weakening the gating signal.

For completeness, a conditional video-diffusion variant of the same task is implemented and documented in Section 5.7; it is treated as an exploratory complement to the ConvGRU pipeline rather than a substitute.

5 Method

5.1 Architecture Overview

Figure 1 shows the unified ConvGRU-based architecture used by all three model variants. Solid arrows denote forward computation; the dashed arrow denotes the uncertainty-gated memory-update path used only by the uncertainty variant. The no-memory baseline removes the entire 3D-memory branch and the memory-encoder/fuse path; the memory variant uses all solid arrows; the uncertainty variant additionally uses the heteroscedastic head and the dashed write-gating loop.

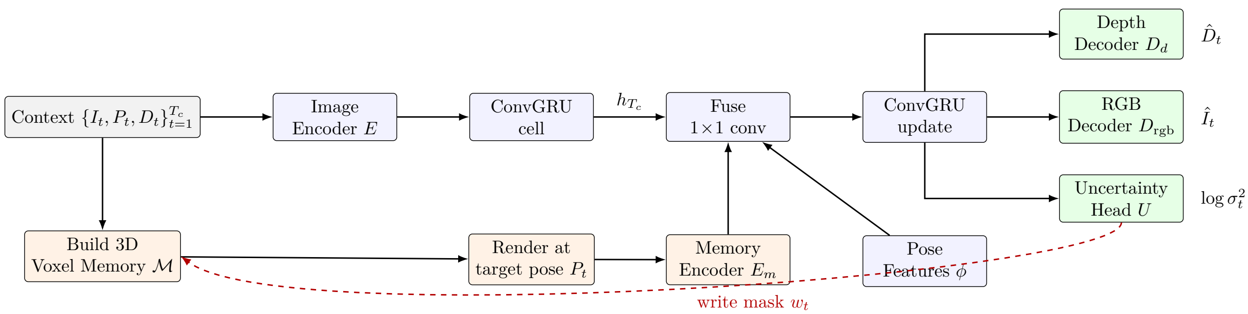

Figure 1: Unified architecture used by all three ConvGRU variants. Top row: image encoder, recurrent ConvGRU cell, fusion block, and prediction heads (RGB, depth, and – for the uncertainty variant – a heteroscedastic head). Bottom row: persistent 3D voxel memory $\mathcal{M}$, target-view render, and memory encoder. The dashed red path is the uncertainty-gated write loop used only by the uncertainty variant.

5.2 Data Pipeline

We export MOVi-A clips into compressed `.npz` shards containing RGB video, depth, camera poses, camera intrinsics, and metadata. A manifest file lists all clip shards and is the only source of truth that downstream training scripts read. From these clips we extract overlapping context/target windows; the number of windows per clip is capped separately for the train and validation splits to control runtime. We use a fixed random seed and a fixed `val_ratio = 0.2` to split clips into train and validation sets, so every model variant sees identical data.

5.3 No-Memory ConvGRU Baseline

Each context frame is encoded by a small convolutional encoder. The resulting feature sequence is passed through a ConvGRU temporal cell [4] that produces a hidden state. The hidden state is then decoded, conditioned on each target pose, into the predicted target RGB sequence. The training loss is a weighted combination of per-pixel L1 reconstruction and a dynamic-region L1 term that up-weights pixels where the target frame moves relative to the last context frame.

This baseline isolates the value of *recurrence alone* on this task and serves as the reference for the memory-augmented models.

5.4 Memory-Conditioned ConvGRU

The memory-conditioned model adds an explicit persistent 3D scene memory. From the context RGB-D and poses we construct a voxelized memory grid of resolution $48 \times 40 \times 48$. For each target pose, this grid is differentiably rendered into target-view-aligned RGB, depth, and a coverage mask. The model receives the context features (as in the baseline) and additionally a rendered memory conditioning tensor for each target frame.

The training objective for this model adds, on top of the baseline RGB and dynamic-region losses, (i) an auxiliary depth L1 loss and (ii) a memory-covered L1 loss that focuses error on regions for which the memory render returned a non-empty pixel. This pushes the model to actually *use* the memory render rather than ignore it.

The memory model is warm-started from the converged no-memory ConvGRU: weights with matching shapes (encoder, ConvGRU cell, RGB decoder, etc.) are copied over so that the memory model only has to learn the new memory pathway, not the entire predictor. This decomposition stabilizes training and, more importantly, makes any metric difference between the no-memory and memory variants directly attributable to the memory pathway.

5.5 Uncertainty-Gated Memory Writes

The uncertainty variant adds a heteroscedastic head [6] that predicts a per-pixel log-variance. During training we add a Gaussian negative log-likelihood term whose weight, write-confidence threshold, and confidence gamma are all exposed as hyperparameters. At rollout time, the predicted confidence is used to *gate* which pixels are written into the persistent memory: low-confidence pixels are down-weighted or skipped, which prevents the memory from being contaminated by uncertain estimates.

The uncertainty model is warm-started from the converged memory-conditioned ConvGRU. We follow the same data and optimization settings as the memory model, plus the additional uncertainty hyperparameters (NLL weight 0.25, write-confidence threshold 0.99, confidence gamma 4.0).

5.6 Formal Specification

This subsection states precisely what every variant computes and the exact loss it minimises, so the entire pipeline is reproducible from equations.

Notation. Let T_c and T_p be context and prediction lengths, $H{=}W{=}64$ the image side, and K the camera intrinsics. For each window we have context frames $I_{1:T_c} \in \mathbb{R}^{T_c \times 3 \times H \times W}$, depths $D_{1:T_c} \in \mathbb{R}^{T_c \times H \times W}$, context poses $P_{1:T_c}$, and target poses $P_{T_c+1:T_c+T_p}$, all in SE(3).

Encoder and recurrence. Each image is encoded by a small convolutional encoder $E : \mathbb{R}^{3\times H\times W} \to \mathbb{R}^{C\times h\times w}$ (with $C = 128$). The context hidden state is computed by a ConvGRU cell [4] iterated over the context:

$$
\begin{aligned}
e_t &= E(I_t), && (1)\\
z_t &= \sigma(W_z * [e_t, h_{t-1}]), && (2)\\
r_t &= \sigma(W_r * [e_t, h_{t-1}]), && (3)\\
\tilde{h}_t &= \tanh(W_h * [e_t, r_t \odot h_{t-1}]), && (4)\\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t, && (5)
\end{aligned}
$$

where $*$ denotes 2D convolution with reset gate r_t and update gate z_t, and $\odot$ is elementwise product. We unroll this for $t = 1, \ldots, T_c$ to obtain h_{T_c}.

Pose features. Given an anchor pose P_a (the last context pose) and a target pose P_t, we form a relative-pose vector

$$\phi(P_a, P_t) = \mathrm{vec}\big((P_a^{-1} P_t)_{[:3,:4]}\big) \ \in \ \mathbb{R}^{12}, \tag{6}$$

and project it through a two-layer MLP, then broadcast to a spatial map of shape $C \times h \times w$.

Persistent 3D memory (memory variants). We build a voxelised memory $\mathcal{M}$ on a regular grid of resolution $48 \times 40 \times 48$. The memory is constructed by unprojecting context RGB-D into world coordinates and splatting:

$$\mathbf{X}_{t,p} = P_t\, K^{-1} \big[u_p, v_p, 1\big]^\top D_t(u_p, v_p), \qquad \mathcal{M} \leftarrow \mathrm{Splat}\big(\{(\mathbf{X}_{t,p}, I_t(u_p, v_p))\}_{t,p}\big). \tag{7}$$

For each target pose P_t the memory is rendered into a target-view conditioning tensor R_t together with a coverage mask $m_t^{\mathcal{M}} \in \{0,1\}^{H \times W}$ (true where $\mathcal{M}$ projects to a non-empty pixel) and a rendered depth. The memory encoder E_m produces a feature map of the same shape as $E(I)$, allowing direct fusion.

Memory-conditioned ConvGRU update. At each prediction step $t \in \{T_c{+}1, \dots, T_c{+}T_p\}$ the model fuses the previous-frame latent, the memory-render latent, and the pose embedding through a $1{\times}1$ convolution and steps the ConvGRU once more:

$$g_t = \mathrm{Conv}_{1\times 1}\big([\, E(\hat{I}_{t-1}),\, E_m(R_t),\, \phi(P_a, P_t)\,]\big), \quad h_t = \mathrm{ConvGRU}(g_t, h_{t-1}), \tag{8}$$

$$\hat{I}_t = \mathrm{clip}_{[0,1]}\big(\hat{I}_{t-1} + \alpha\, \tanh\big(D_{\mathrm{rgb}}(h_t)\big)\big), \qquad \hat{D}_t = \mathrm{ReLU}\big(D_d(h_t)\big), \tag{9}$$

with residual scale $\alpha = 0.25$. We start from $\hat{I}_{T_c} := I_{T_c}$. The no-memory baseline is the same recurrence with $E_m(R_t)$ removed and a sum-fusion of $E(\hat{I}_{t-1})$ and the pose map.

Heteroscedastic head and write gating. The uncertainty variant adds a head U that predicts a per-pixel log-variance, clipped for stability:

$$\log \sigma_t^2 \ = \ \mathrm{clip}\big(U(h_t),\, \ell_{\min},\, \ell_{\max}\big). \tag{10}$$

A confidence map and a binary write mask are derived as

$$c_t = \frac{e^{-\log \sigma_t^2}}{1 + e^{-\log \sigma_t^2}}, \qquad \tilde{c}_t = c_t^{\gamma}, \qquad w_t = \mathbb{1}[\tilde{c}_t \geq \tau], \tag{11}$$

with confidence-gamma $\gamma = 4$ and write-confidence threshold $\tau = 0.99$. Memory writes during rollout use w_t as a gating mask so that low-confidence pixels do not contaminate $\mathcal{M}$.

Dynamic and memory-covered masks. Two evaluation/training masks are derived from the inputs:

$$m_t^{\mathrm{dyn}}(u,v) = \mathbb{1}\big[\tfrac{1}{3} \textstyle\sum_c |I_t(c,u,v) - I_{T_c}(c,u,v)| > \theta\big], \tag{12}$$

$$m_t^{\mathcal{M}}(u,v) = (\text{memory-render coverage mask}), \tag{13}$$

with motion threshold $\theta = 0.03$.

Composite training loss. For a window we minimise

$$\mathcal{L}_{\text{rgb}} = \tfrac{1}{T_p HW} \textstyle\sum_{t,c,u,v} |I_t - \hat{I}_t|, \tag{14}$$

$$\mathcal{L}_{\text{dyn}} = \text{MaskedL1}(\hat{I}, I; m^{\text{dyn}}), \tag{15}$$

$$\mathcal{L}_{\text{depth}} = \text{MaskedL1}(\hat{D}, D; \mathbb{1}[D > 0]), \tag{16}$$

$$\mathcal{L}_{\mathcal{M}} = \text{MaskedL1}(\hat{I}, I; m^{\mathcal{M}}), \tag{17}$$

$$\mathcal{L}_{\text{NLL}} = \tfrac{1}{T_p HW} \textstyle\sum_{t,u,v} \Big[\tfrac{1}{2} e^{-\log \sigma_t^2}\, \bar{\delta}_t^2 + \tfrac{1}{2} \log \sigma_t^2\Big], \quad \bar{\delta}_t^2 = \tfrac{1}{3} \textstyle\sum_c (\hat{I}_t - I_t)^2, \tag{18}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rgb}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \lambda_{\mathcal{M}} \mathcal{L}_{\mathcal{M}} + \lambda_{\text{NLL}} \mathcal{L}_{\text{NLL}}. \tag{19}$$

We use $\lambda_{\text{dyn}} = 3$, $\lambda_{\text{depth}} = 0.1$ (memory variants), $\lambda_{\mathcal{M}} = 1$ (memory variants), and $\lambda_{\text{NLL}} = 0.25$ (uncertainty variant). The no-memory baseline keeps only $\mathcal{L}_{\text{rgb}}$ and $\mathcal{L}_{\text{dyn}}$.

Evaluation metrics. For evaluation we additionally compute $\text{PSNR} = -10 \log_{10} \text{MSE}$, the dynamic and memory-covered L1 above, and the per-window correlation between predicted uncertainty and per-pixel error,

$$\rho_{u,e} \;=\; \text{corr}\big(\log \sigma_t^2,\, \bar{\delta}_t^2\,\big), \tag{20}$$

together with a write-coverage statistic $\mathbb{E}[w_t]$.

5.7 Diffusion Variant (Exploratory)

For completeness, we additionally implemented no-memory and memory-conditioned diffusion variants of the same task using a small conditional video diffusion model. Configurations matched the ConvGRU pipeline as closely as possible (same data manifest, image size, T_c, T_p). The diffusion variant is intentionally exploratory: as discussed in Section 4, combining a generative sampler with the question “does persistent memory help?” makes attribution of metric changes harder, because metric movement could come either from improved memory conditioning or from the sampler itself. The diffusion runs in our setup did not reach a stable, usable quality regime suitable for like-for-like comparison against the deterministic ConvGRU pipeline; we therefore do *not* report diffusion numbers in our main quantitative tables. The scripts and exact configurations remain in the repository so that the diffusion experiments are fully reproducible.

6 Experimental Setup

Dataset and split. We use 200 MOVi-A clips at 128×128 source resolution, split with a fixed seed into 160 train clips and 40 validation clips. From these we extract up to 16 train and 8 validation context/target windows per clip, yielding 2,560 training windows and 320 validation windows.

Architecture. Each ConvGRU model uses `hidden_channels` = 128. Memory-conditioned variants use a memory grid of $(48, 40, 48)$, memory stride 1, and splat radius 1.

Training schedule. The no-memory and memory models are trained for 50 epochs each with batch size 16. The uncertainty variant is trained for up to 40 epochs warm-starting from the memory checkpoint. We use Adam with a learning rate of 10^{-3} and report *best validation* checkpoints chosen by the model L1 metric.

Loss configuration.

- Dynamic-region weight: 3.0 (motion threshold 0.03).
- Depth loss weight (memory variants): 0.1.
- Memory-covered loss weight (memory variants): 1.0.
- Memory render loss weight (memory variants): 0.0.
- Uncertainty NLL weight (uncertainty variant): 0.25, with write-confidence threshold 0.99 and confidence gamma 4.0.

Hardware and runtime. All reported numbers come from single-GPU training on a workstation-class GPU and an L40S-class GPU. The dominant runtime cost is per-sample memory conditioning (voxel write + render), which significantly increases the cost of each training and validation pass for the memory-conditioned models relative to the no-memory baseline.

7 Evaluation Protocol

We evaluate all three trained variants under a unified harness that loads each model's best validation checkpoint and computes metrics on identical validation windows. The reported metrics are:

- **Val L1** (↓): mean RGB L1 over all target pixels.
- **Dyn L1** (↓): RGB L1 restricted to dynamic pixels (target differs from last context frame).
- **Mem-cov L1** (↓): RGB L1 restricted to regions where the oracle memory render is non-empty.
- **PSNR** (↑): mean PSNR over target frames.
- **Unc-corr**: Pearson correlation between predicted uncertainty and per-pixel error (uncertainty model only).
- **Write-cov**: fraction of target pixels that pass the uncertainty write gate (uncertainty model only).

In addition to overall metrics, we report metrics on four diagnostic slices automatically built from the validation set: *high motion* (top-25% by motion fraction), *occlusion recovery* (objects that go from visible to hidden to visible), *high memory coverage* (top-25% by oracle memory render coverage), and *depth-edge heavy* (top-25% by normalized depth-edge fraction).

8 Results

8.1 Overall Quantitative Comparison

Table 1 summarizes the unified validation results from a single evaluation run over all three models on the same 320 validation windows.

Analysis.

- The *memory ConvGRU* improves over the *no-memory* baseline on every reported axis: −12.1% on overall validation L1, −13.0% on dynamic L1, −11.6% on memory-covered L1, and +1.11 dB PSNR.

Model	Val L1 ↓	Dyn L1 ↓	Mem-cov L1 ↓	PSNR (dB) ↑	Unc-corr	Write-cov
ConvGRU no-memory	0.0214	0.1023	0.0653	25.57	—	—
ConvGRU memory	0.0188	**0.0890**	0.0577	26.68	—	0.000
ConvGRU memory + uncertainty	**0.0185**	0.0899	**0.0443**	**26.71**	0.150	0.973

Table 1: Overall validation results on the MOVi-A subset200 split (320 validation windows). Lower is better for L1 metrics, higher is better for PSNR. Numbers from `outputs/eval_final/evaluation.json`.

- The *uncertainty* variant matches or improves the memory model on every metric except a marginal 0.001 increase in dynamic L1, while reducing memory-covered L1 from 0.0577 to 0.0443 (−23.2%) – the metric most directly tied to how faithfully the memory pathway is used.
- The predicted log-variance is positively correlated with per-pixel prediction error (r=0.150). Although not formally calibrated, the signal is meaningfully predictive of regions where the model is uncertain, which is the property the gating mechanism relies on.
- A write-coverage of 0.973 at the chosen threshold indicates that the gate is selective without preventing the persistent memory from being updated, which is a desirable operating point at this scale.

8.2 Slice-Level Comparison

Table 2 reports validation L1 for each model across diagnostic slices. The same ranking (*uncertainty* $\leq$ *memory* $<$ *no-memory*) holds on every non-empty slice, and the uncertainty variant is best on all non-trivial slices.

Slice (count)	No-memory	Memory	Memory + uncertainty
all (320)	0.0214	0.0188	**0.0185**
high_motion (80)	0.0295	0.0263	**0.0260**
occlusion_recovery (37)	0.0232	0.0205	**0.0203**
high_memory_coverage (80)	0.0287	0.0259	**0.0256**
depth_edge_heavy (80)	0.0276	0.0246	**0.0243**

Table 2: Validation L1 (↓) per slice. The *high_camera_motion* slice was empty on this MOVi-A subset because the camera is effectively static and is therefore omitted.

8.3 Qualitative Comparison Across Models

Figure 2 shows the unified comparison panel for the median validation case. Each panel stacks the context strip, the ground-truth target strip, the no-memory prediction, the memory prediction, the uncertainty prediction, and (for the uncertainty model) the predicted uncertainty map and derived write mask, all on the same target window.

Across all four panel cases, the memory and uncertainty variants preserve more high-frequency object detail and track moving objects more cleanly than the no-memory baseline, consistent with their lower dynamic-region and memory-covered L1 in Tables 1 and 2. The uncertainty maps in the bottom rows concentrate on object boundaries and recently-disoccluded regions, which is exactly where one would want to suppress memory writes to avoid contaminating the persistent memory with noisy estimates.

8.4 Uncertainty-Specific Diagnostics

Figure 6 isolates the uncertainty output and the derived binary write mask from a representative validation window. The pattern matches expectations: low confidence at motion boundaries and around fine object edges; the write mask suppresses memory updates exactly there.

8.5 Diffusion Outcome

Consistent with the rationale in Section 4, we keep the diffusion variant out of the head-to-head quantitative comparison. Mixing a generative sampler into the same table would conflate the variable of interest (the memory pathway) with sampling-procedure variability, which is exactly what we wanted to avoid by choosing a deterministic backbone. We retain the training scripts and configurations in the repository so that the diffusion methodology is documented end-to-end and can be re-run for follow-up work.

9 Discussion

Persistent memory yields measurable gains at student scale. The principal empirical observation is that explicit persistent memory provides consistent, attributable improvements over a matched no-memory ConvGRU at the same training budget. The gains are most pronounced on dynamic-region L1 and on memory-covered L1, which is consistent with the hypothesis that memory contributes most in regions where the last-frame baseline is weakest – namely moving and recently-disoccluded content.

Uncertainty gating sharpens the memory pathway. The uncertainty variant preserves the gains of the memory model on overall L1 and PSNR, and additionally reduces memory-covered L1 by 23.2%, the metric most directly tied to the fidelity of the persistent memory itself. The predicted log-variance is positively correlated with per-pixel error (r=0.150), suggesting that even without formal calibration the signal localises uncertain regions accurately enough to be useful as a gating mask.

Sequential warm-starting stabilises training. Training the memory model from random weights produced markedly less stable optimisation than warm-starting from the converged no-memory checkpoint, and the same pattern held for warm-starting the uncertainty model from the memory checkpoint. Sequential warm-starting effectively decomposes the optimisation problem into "learn a recurrent video predictor" followed by "learn to use the memory pathway," which we found to be an effective design pattern at this scale.

Methodological coherence of the recurrent backbone. The empirical findings reinforce the rationale of Section 4: because the no-memory and memory variants share an identical predictor and the uncertainty head is itself deterministic, every metric movement we report is attributable to the memory pathway and to the gating mechanism specifically. The diffusion variant, while a valid generative complement, addresses a different research question.

10 Limitations

- **Replication scope.** This work intentionally replicates only the memory mechanism of the reference architecture [1], not the full production-scale training pipeline; reported metrics are comparative under matched settings rather than absolute state-of-the-art.

- **Spatial resolution.** Training at 64×64 is sufficient to rank model variants reliably but is too low to support photorealistic qualitative claims.
- **Uncertainty calibration.** The predicted log-variance is positively correlated with prediction error but is not formally calibrated; a post-hoc temperature-scale or isotonic calibration step is left as future work.
- **Camera motion.** The MOVi-A subset used here has effectively static cameras, so the `high_camera_motion` slice is empty by construction. Evaluating on a dataset variant with non-trivial camera motion would test whether the persistent-memory gains generalise beyond object-motion regimes.

11 Conclusion and Future Work

We have presented a student-scale, end-to-end replication of the persistent-memory mechanism for 3D world modeling on the MOVi-A benchmark. Across overall validation metrics and four diagnostic slices, the memory-conditioned ConvGRU consistently improves upon a matched no-memory baseline, and an uncertainty-gated extension further reduces memory-covered error by 23% while producing a predicted-uncertainty signal that positively correlates with per-pixel error. The deterministic recurrent backbone is shown, both methodologically and empirically, to be a suitable carrier for isolating the contribution of the memory pathway under a controlled experimental protocol.

Several directions follow naturally from this work. First, the predicted uncertainty is informative but uncalibrated; post-hoc calibration is expected to tighten the connection between confidence and reliability of memory writes. Second, the present study is conducted with effectively static cameras; extending the evaluation to a benchmark with camera motion would verify whether the gains generalise to viewpoint-induced occlusion. Third, the exploratory diffusion variant could be revisited with a sampler designed to interact with persistent memory directly, which would allow a fair like-for-like comparison against the recurrent backbone studied here.

Reproducibility

- Data preparation and environment setup are documented in `README.md` of the project repository.
- Training is launched with `scripts/train_convgru_all.sh` (the full ConvGRU pipeline) or with the convenience wrapper `scripts/train_full_pipeline.sh`.
- Diffusion attempts use `scripts/train_diffusion_all.sh`.
- Unified evaluation is reproduced by `scripts/eval_all.py` with the run mapping used in Section 8; the output of that command is `outputs/eval_final/evaluation.json` and `outputs/eval_final/evaluation.md`.
- Multi-model comparison panels are reproduced by `scripts/make_visual_comparison_panel.py`; the figures in this report are stored under `outputs/report_figures/`.

References

[1] *Learning 3D Persistent Embodied World Models*. Reference manuscript, 2025. (Project reference paper used as the methodological starting point for this work.)

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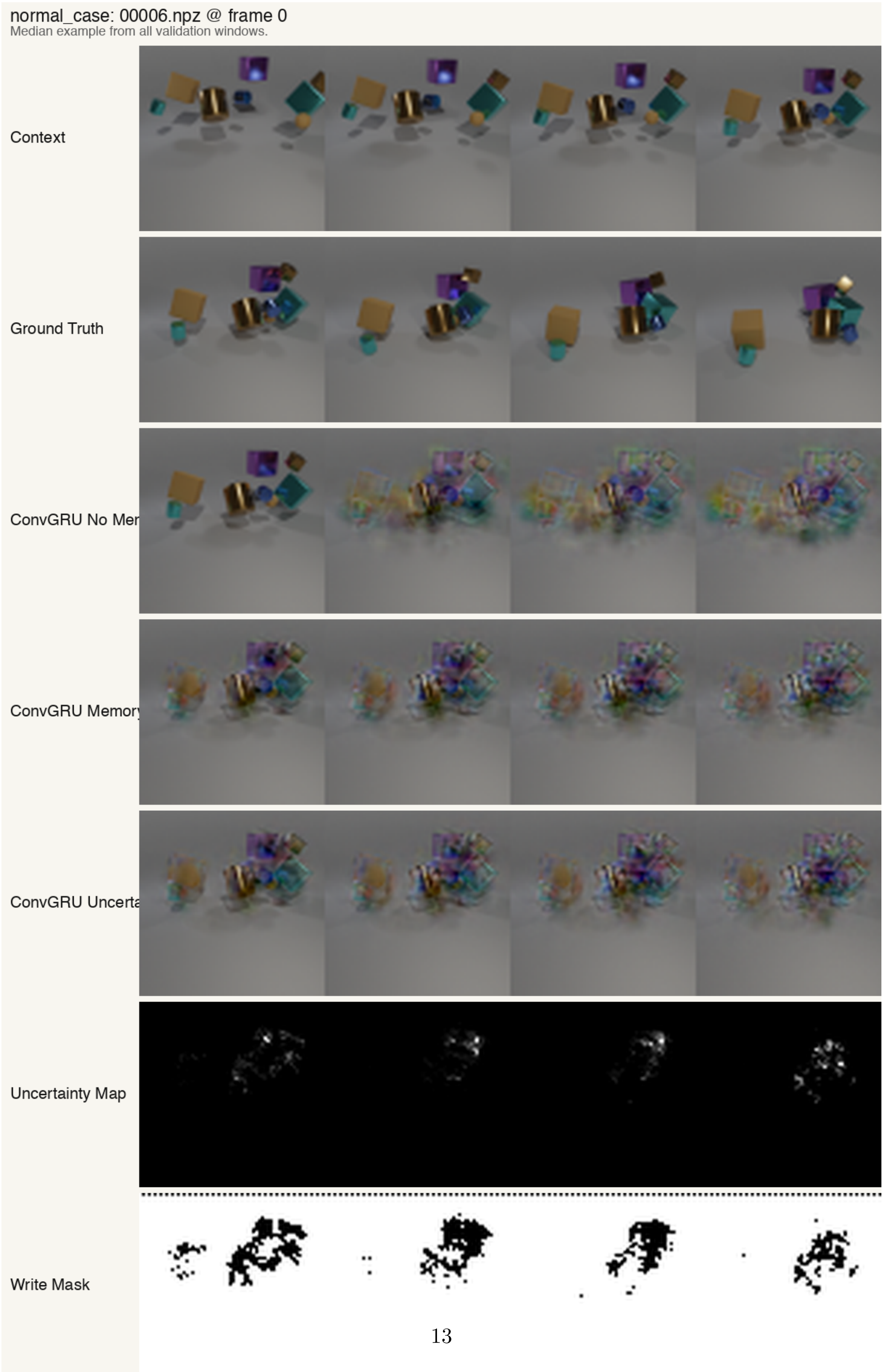

Figure 2: Multi-model comparison on the median validation case. The memory and uncertainty

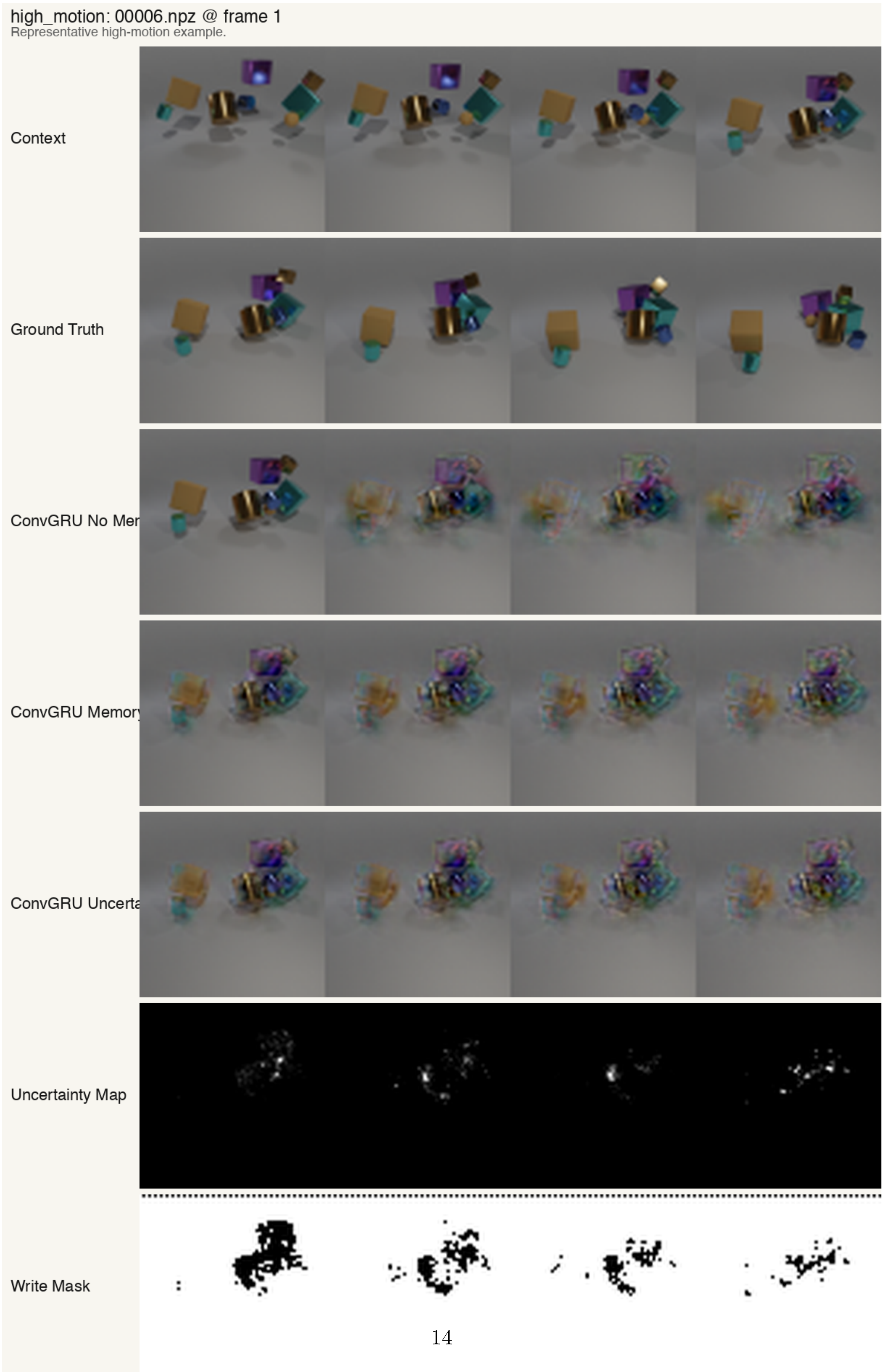

Figure 3: Multi-model comparison on a high-motion validation case.

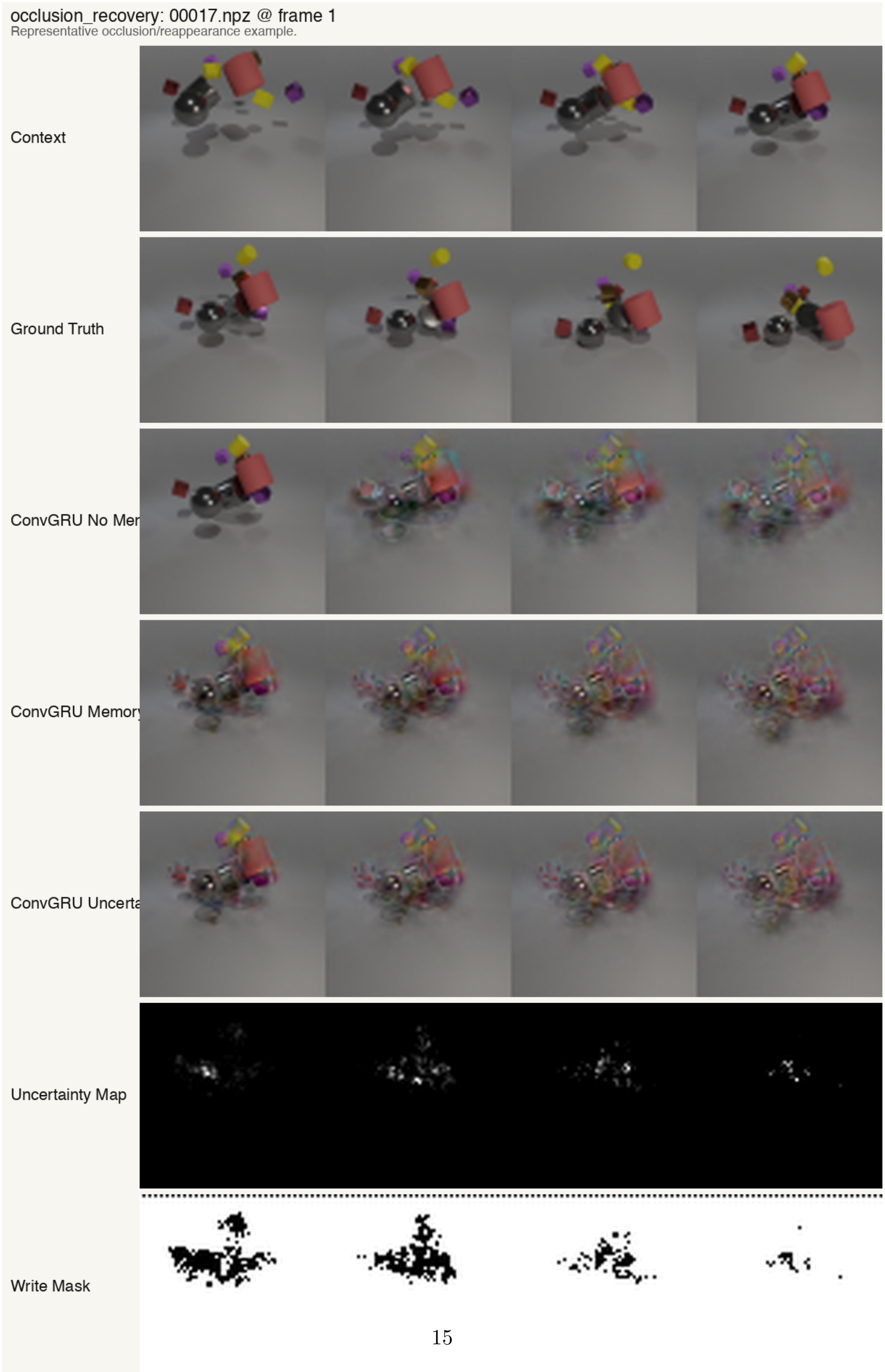

Figure 4: Multi-model comparison on an occlusion-recovery validation case (an object that becomes

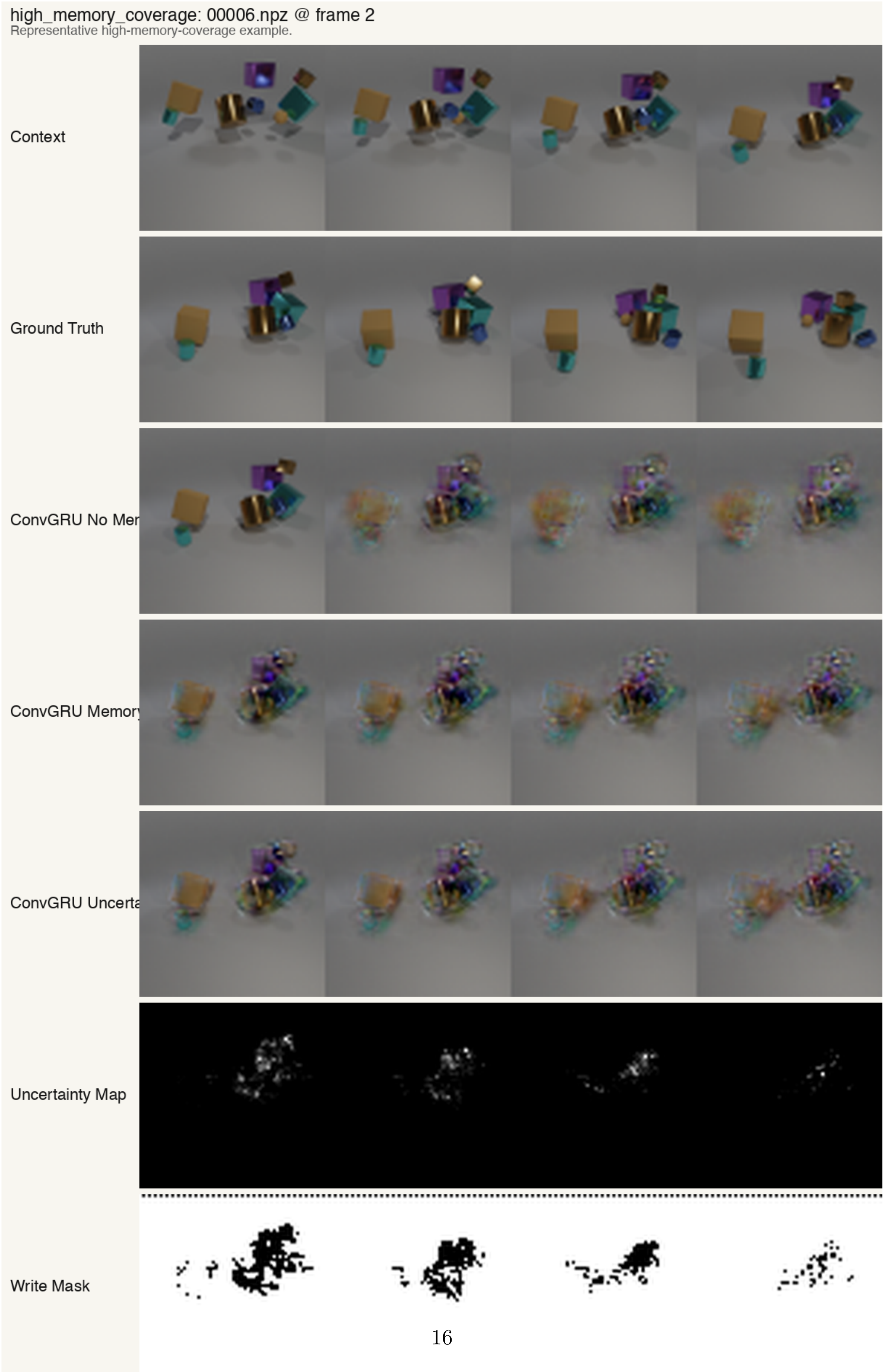

Figure 5: Multi-model comparison on a high-memory-coverage validation case.

Figure 6: Uncertainty diagnostics. Left: predicted per-pixel confidence (brighter = more confident). Right: derived write mask (white = allowed to update memory).